

**Smt. Chandaben Mohanbhai Patel Institute of Computer Applications,
CHARUSAT
BCA Semester-III**

CAUE302 : Python Programming

Practical 1: Introduction to Python Programming Environment

1. Introduction to Python

Python is a high-level, interpreted, object-oriented, and general-purpose programming language developed by **Guido van Rossum** in 1991.

Python is widely used in:

- Web Development
- Data Science
- Artificial Intelligence
- Machine Learning
- Automation
- Scientific Computing

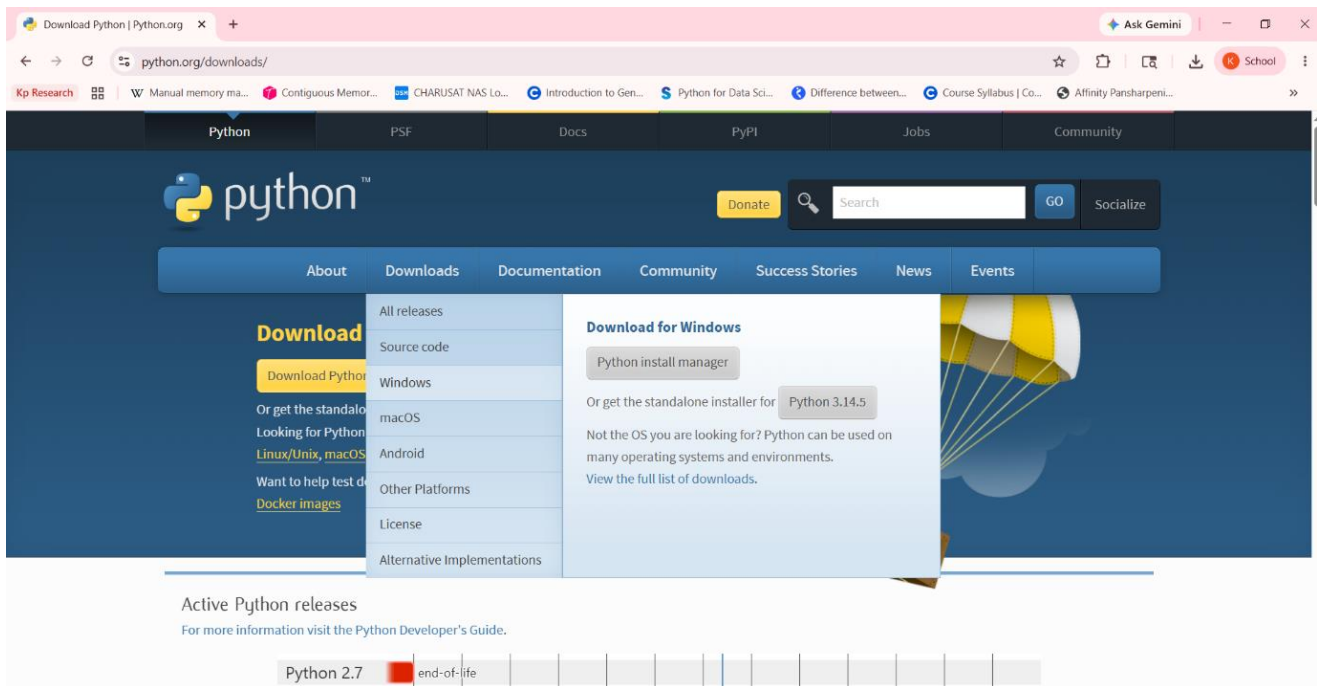
Features of Python

- Easy to Learn
- Simple Syntax
- Platform Independent
- Open Source
- Interpreted Language
- Large Standard Library

2. Downloading and Installation of Python

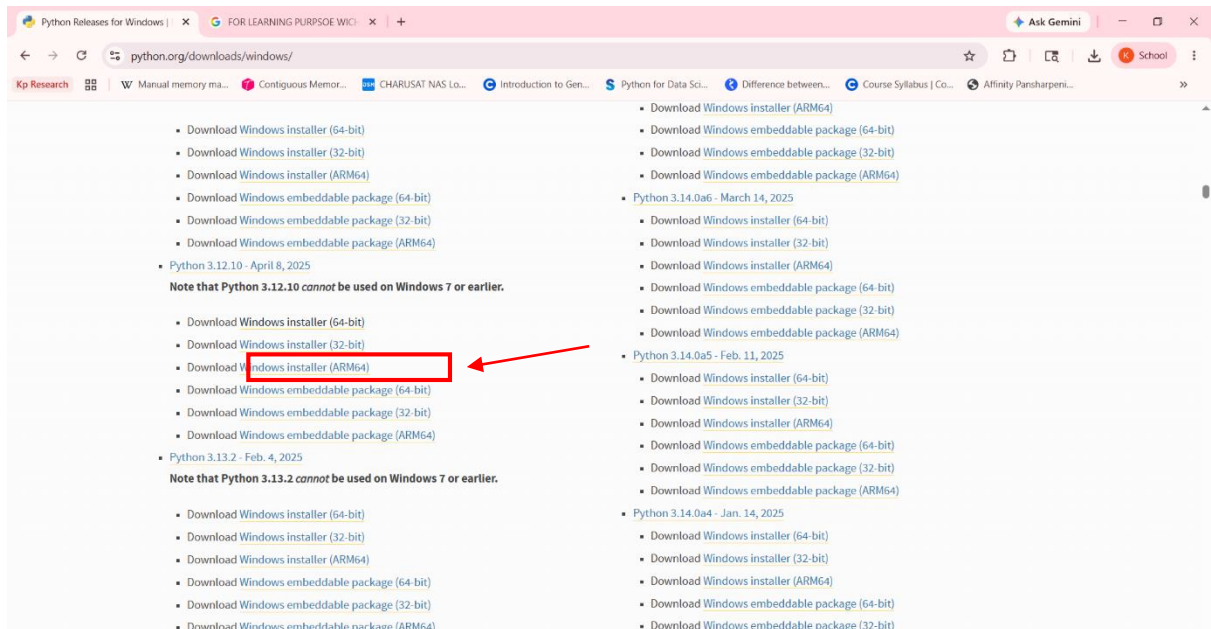
Step 1: Download Python

<https://www.python.org/downloads/>



Step 1: Download Python

- Scroll down to "Looking for a specific release?"



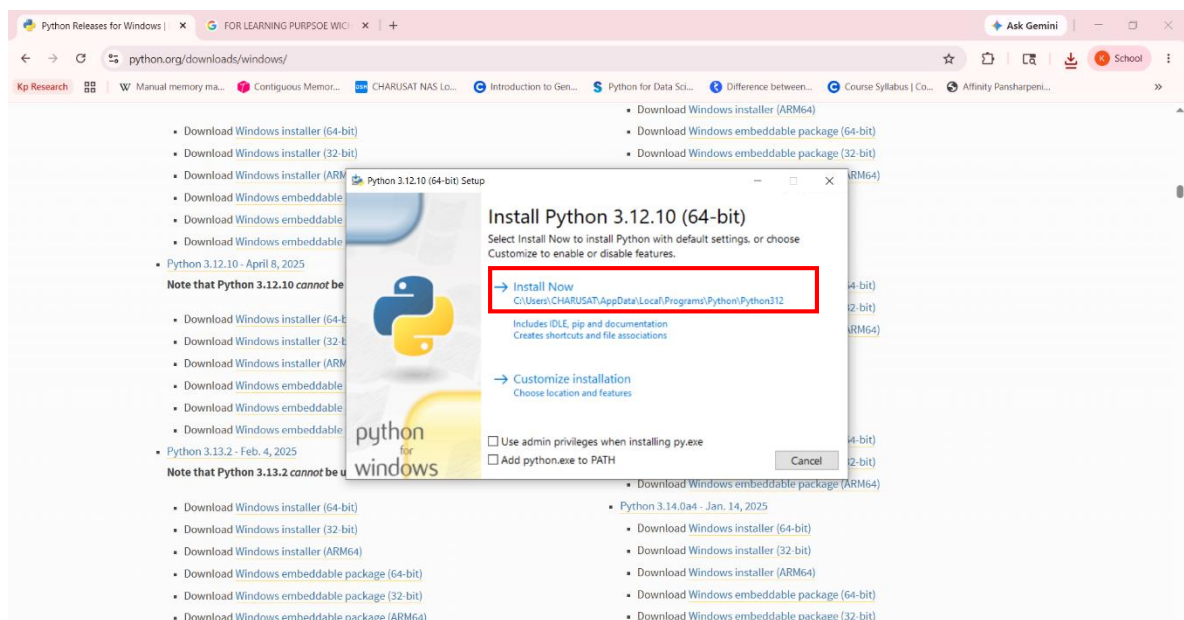
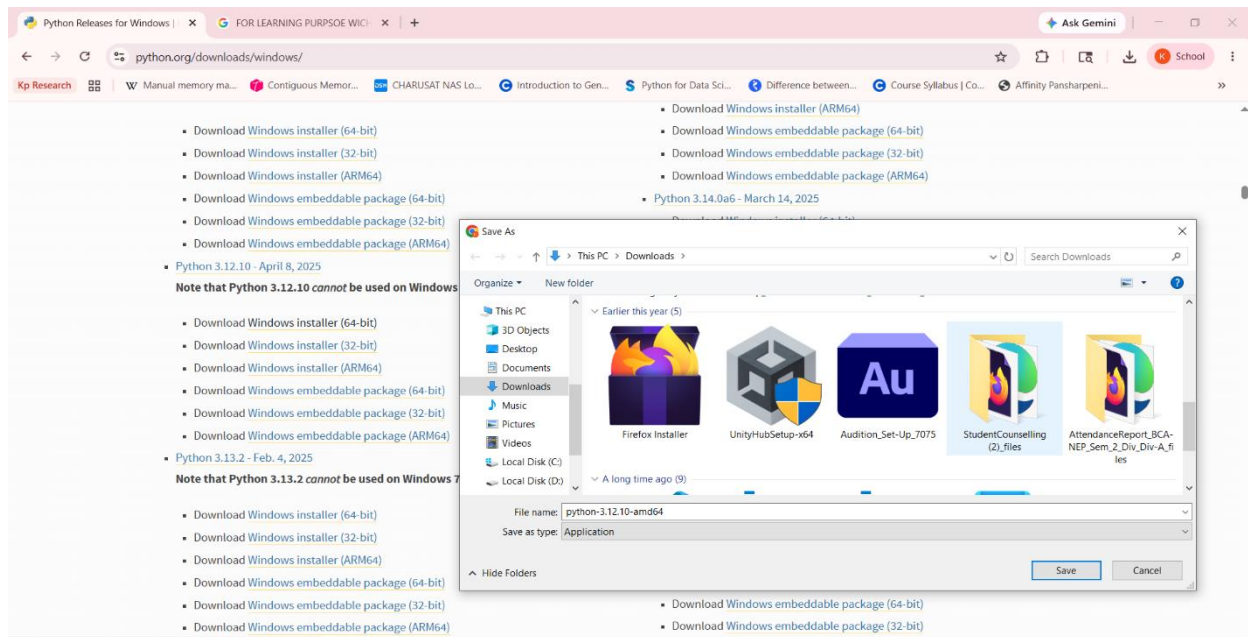
- Download **Python 3.12.10** for Windows.

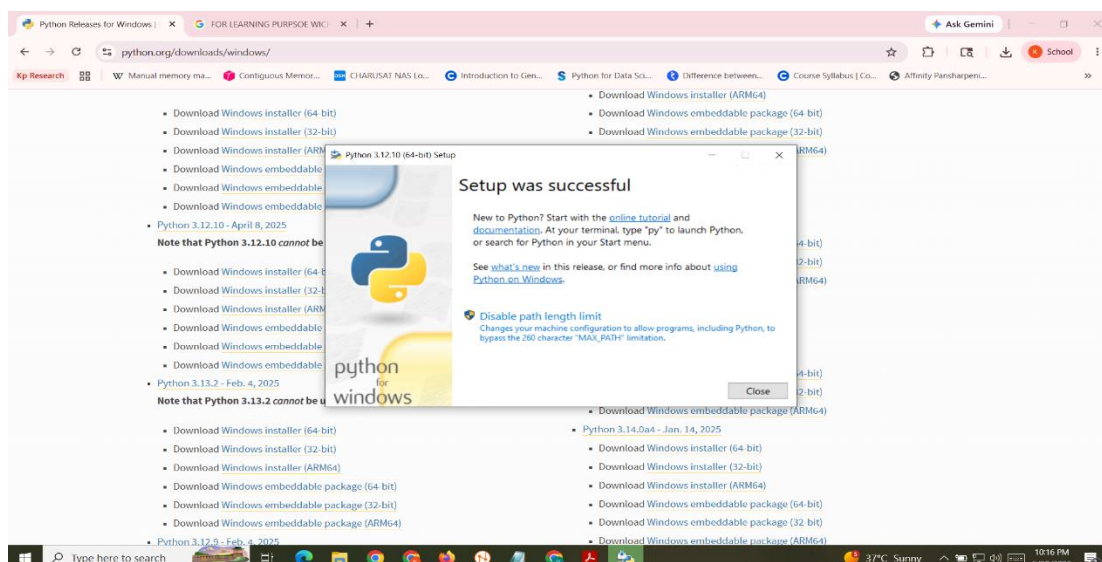
Why Python 3.12.10?

For this course, students should install **Python 3.12.10** to ensure that all students use the same version during practical sessions and avoid compatibility issues.

Step 2: Install Python 3.12.10

1. Double-click the downloaded installer.





- Wait for the installation to complete.
- Click **Close**.

Step 3: Verify Installation

Open Command Prompt and type:

```
C:\Users\CHARUSAT>python --version
Python was not found; run without arguments to install from the Microsoft Store, or disable this shortcut from Settings > Apps > Advanced app settings > App e
```

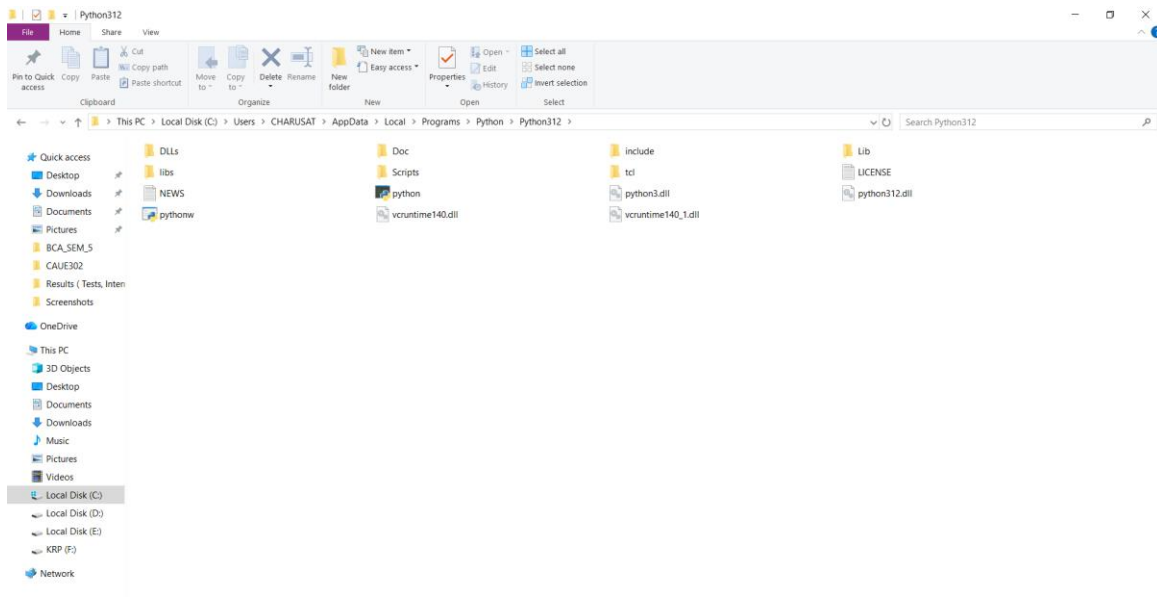
This error means Python is installed, **but** Windows Command Prompt cannot find the Python executable in the system PATH.

Check if Python is in PATH:

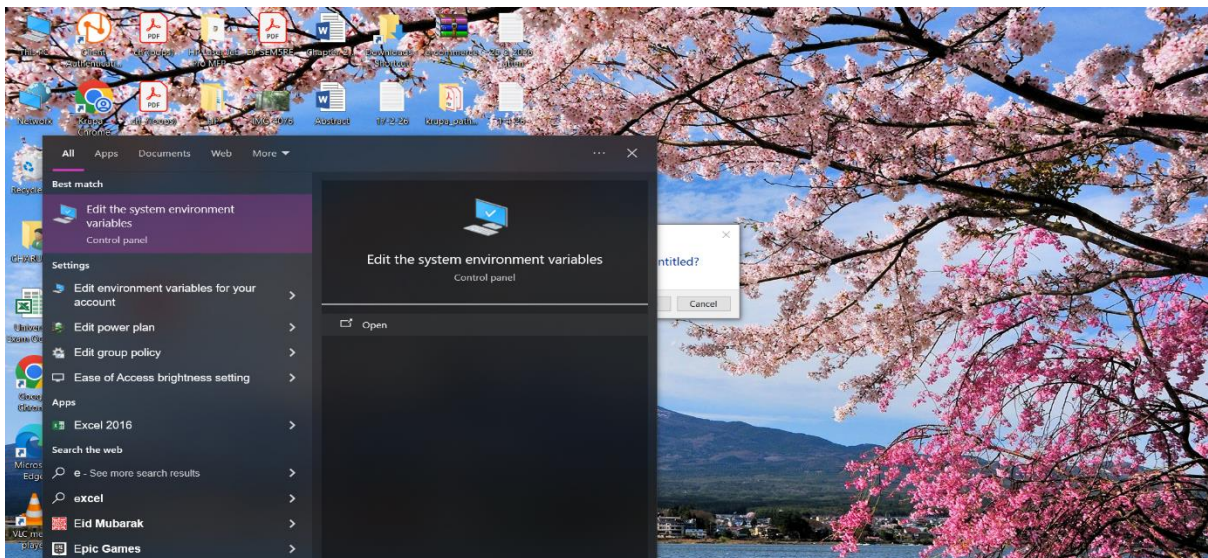
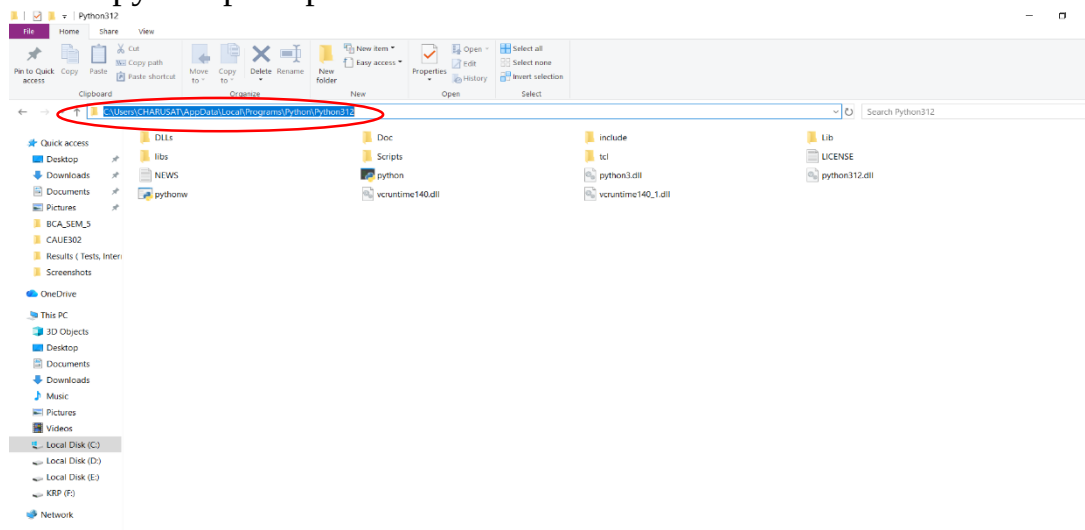
Open Command Prompt and run:

where python

<c:\Users\CHARUSAT\AppData\Local\Programs\Python\Python312\python.exe>

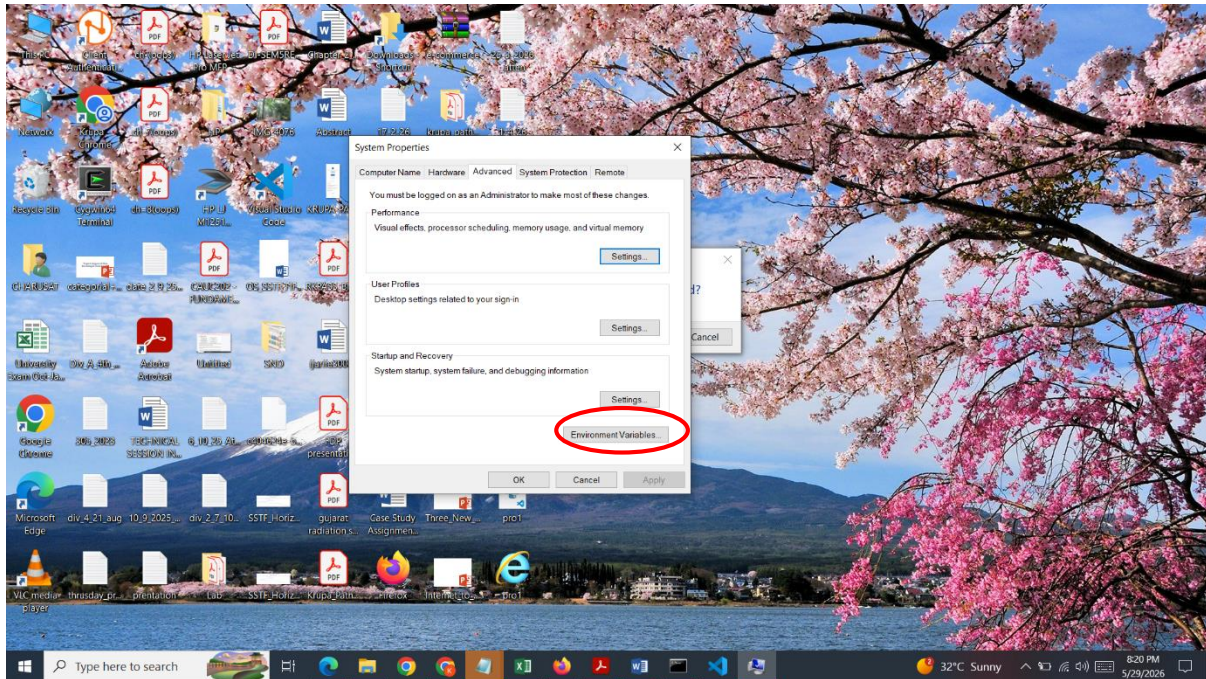


Copy complete path:

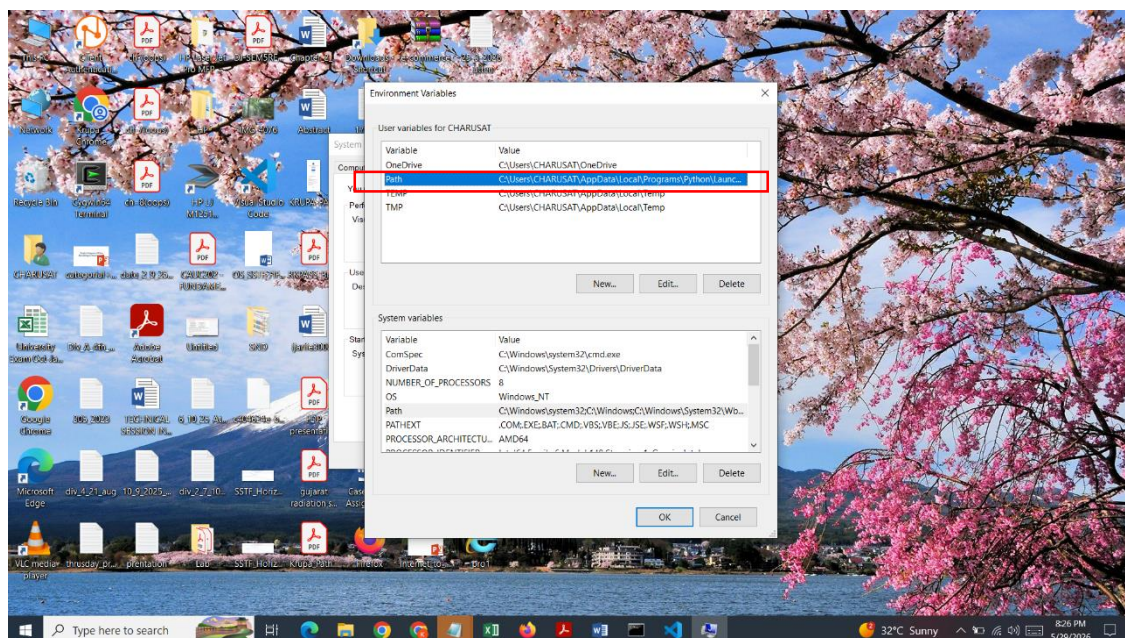


Write Edit the System Environment Variables in search bar and open it.

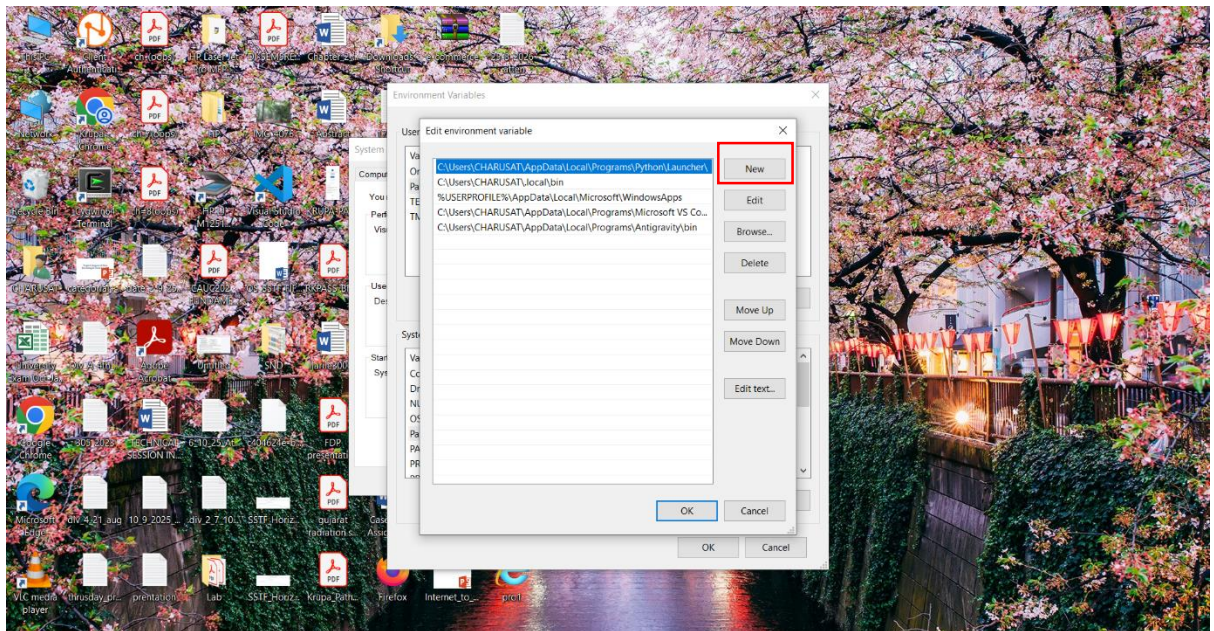
Click on Environment Variables



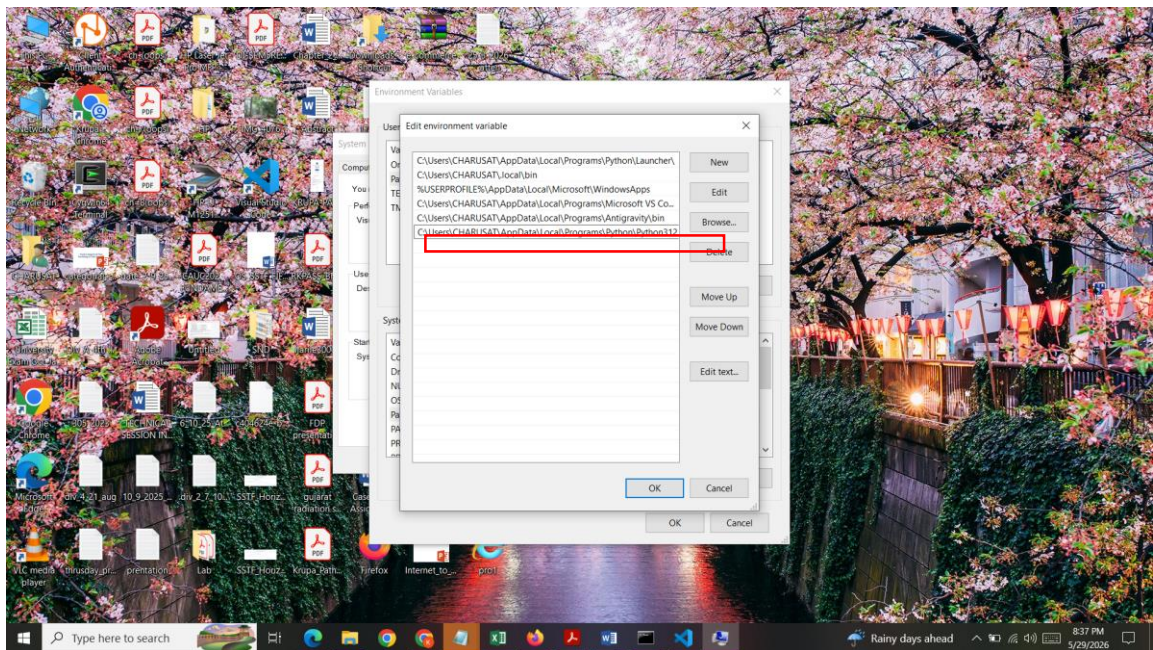
- Under **User variables for CHARUSAT**
- Select **Path**



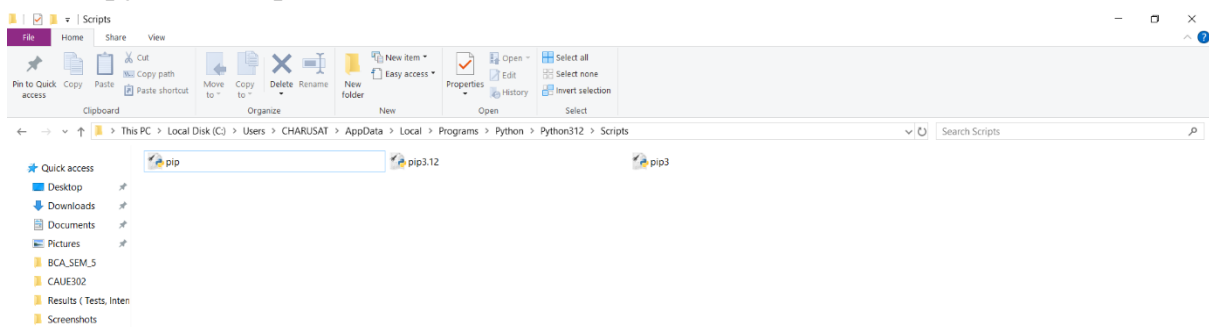
- Click **Edit**



- Click on new Button add path.

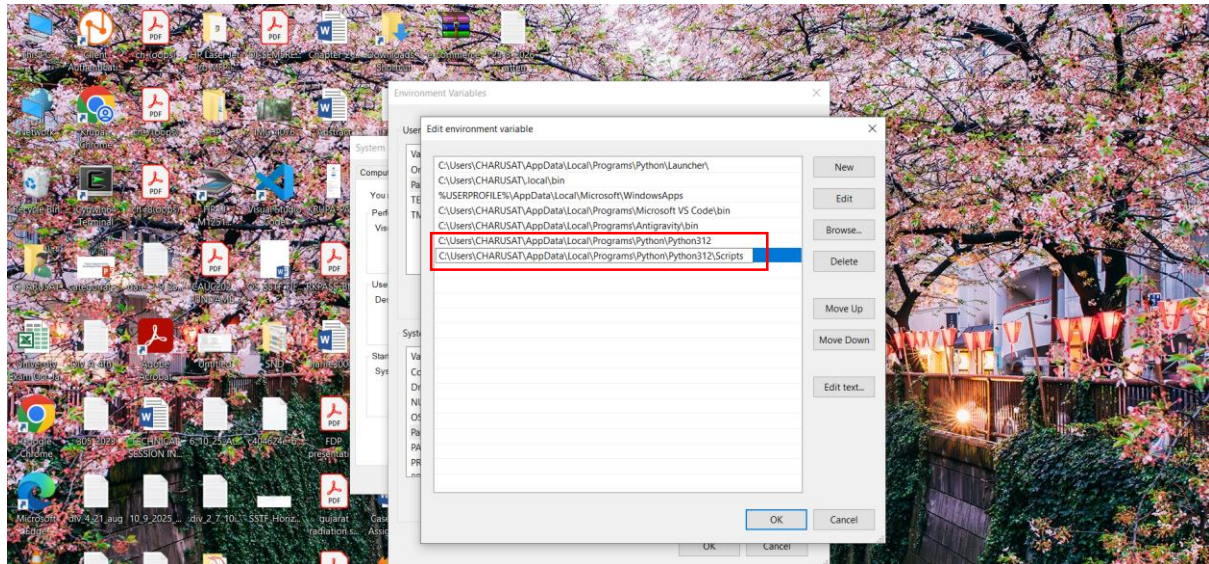


- Copy another path:



Click on new again and paste copied path:

Ok.



```

Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

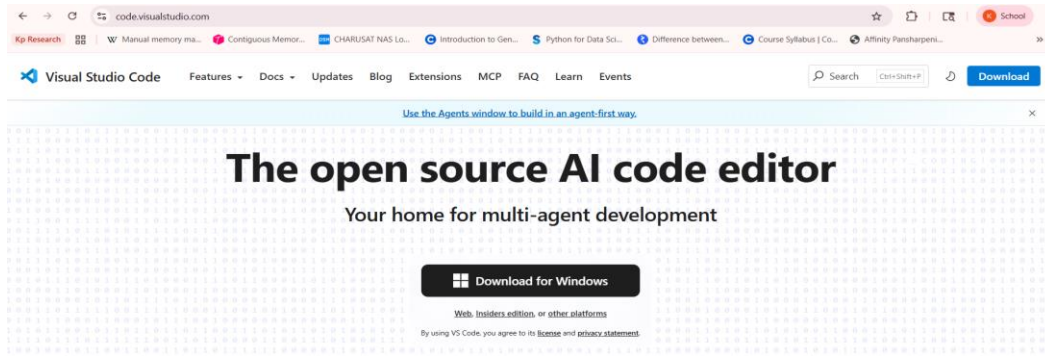
C:\Users\CHARUSAT>python --version
Python 3.12.10

C:\Users\CHARUSAT>
  
```

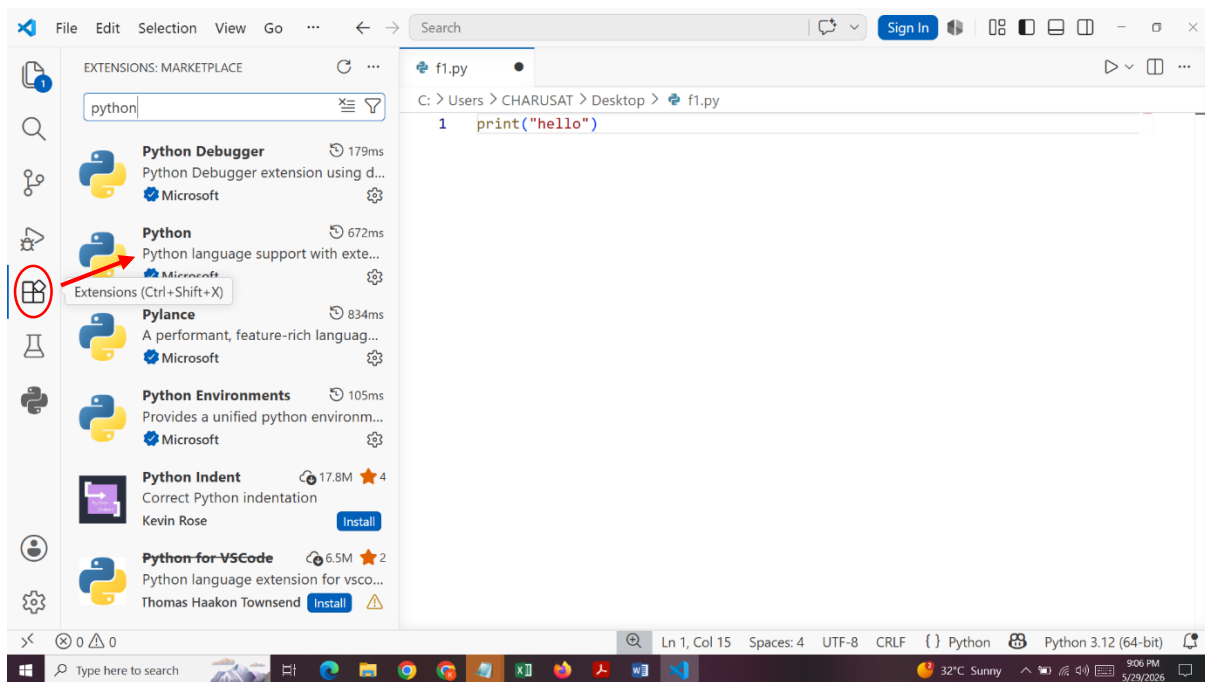
3. Installation of Visual Studio Code (VS Code)

What is VS Code?

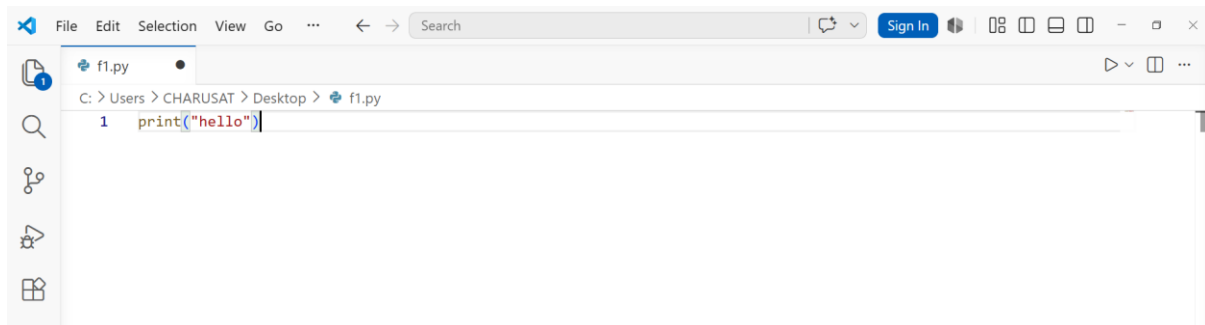
Visual Studio Code (VS Code) is a free and lightweight source code editor developed by Microsoft. It is widely used for Python programming because it provides features such as syntax highlighting, code completion, debugging, and extension support.



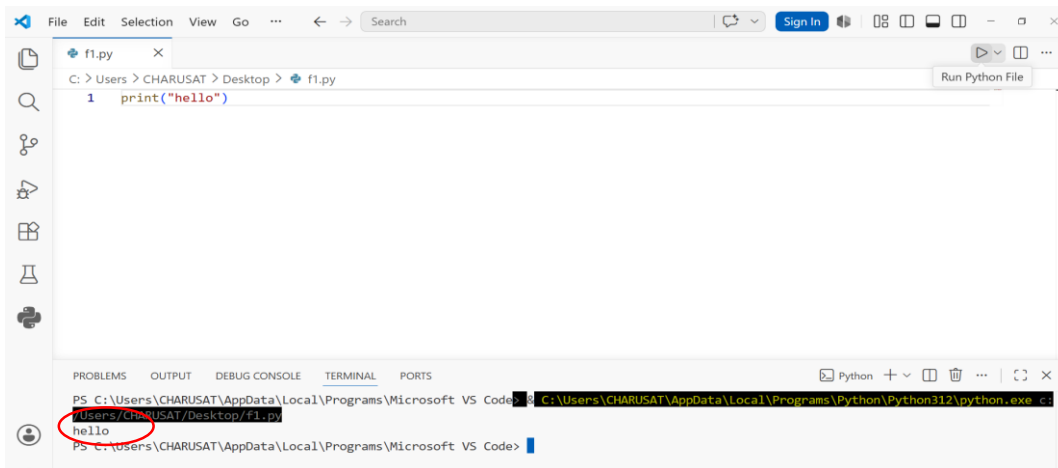
Installing Python Extension in VS Code



Verify Python in VS Code



Click on run button.



Installation of Jupyter Notebook Extension in VS Code

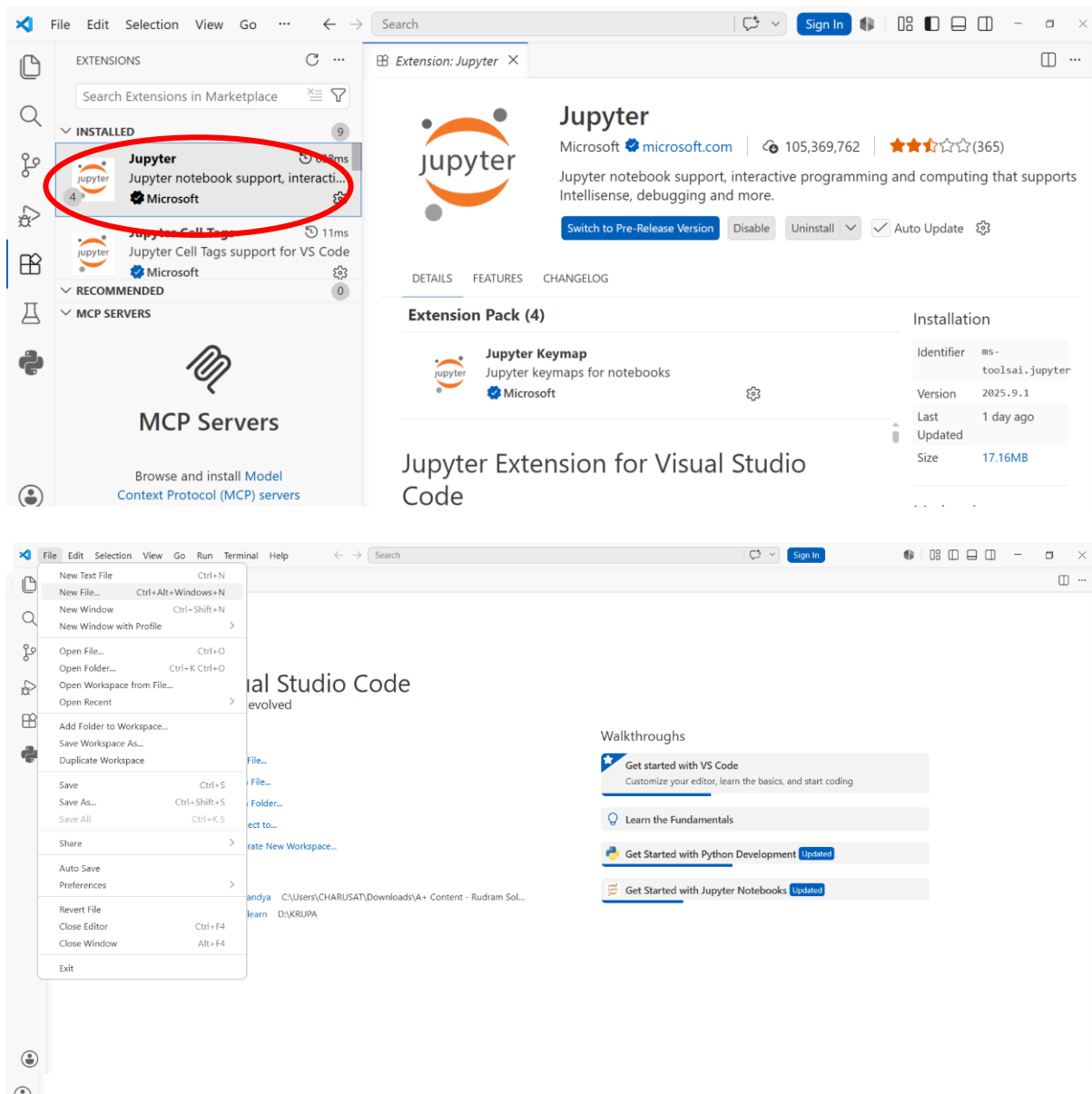
What is Jupyter Notebook?

A **Jupyter Notebook** is an interactive environment used to write and execute Python code in small sections called **cells**. It is widely used for:

- Learning Python
- Data Science
- Machine Learning
- Data Analysis
- Visualization

Jupyter allows users to write code, execute it, and view the output immediately.

Installing the Jupyter Extension



Practical 1: Creating and Running Your First Python Program in Jupyter Notebook (VS Code)

Step 1: Open VS Code

- Double-click the VS Code icon.
- You will see the VS Code window.

Step 2: Create a Jupyter Notebook

- Click:
 - File → New File

OR

- Press:
 - Ctrl + N
- Save the file as:
 - FirstProgram.ipynb

What is .ipynb?

- **Interactive Python Notebook**

It allows us to:

- Write code in cells
 - Execute code cell by cell
 - See output immediately
-

Step 3: Understanding the Notebook Interface

- Students will see :+ Code Click it.
 - A code cell will appear.
 - A cell is simply an area where we write Python code.
-

Step 4: Write Your First Python Program

Type:

```
print("Hello World")
```

Step 5: Understanding the Program

What is print()?

- print() is a built-in Python function.
- It displays output on the screen.

- Example:
 - `print("Hello World")`
 - Output:
 - Hello World
-

What are the quotation marks?

- "Hello World"
- This is called a String.
- A String is text enclosed within quotes.

Examples:

```
print("Python")  
print("BCA")  
print("Welcome Students")
```

Step 6: What is a Kernel?

Definition

A Kernel is the engine that executes Python code.

When we click Run:

1. Code goes to Kernel
 2. Kernel executes code
 3. Result is displayed
-

Step 7: Selecting Kernel

At the top-right corner:

Select Kernel

Click it.

Choose:

Python 3.x.x

Example:

Python 3.12.10

After selecting the kernel, the notebook is ready.

Step 8: Running the Program

Method 1

Click:

► Run

beside the cell.

Method 2 (Keyboard Shortcut)

Press:

Shift + Enter

This is the MOST IMPORTANT shortcut.

Remember this shortcut because we will use it throughout the semester.

What Happens After Running?

Input:

```
print("Hello World")
```

Output:

Hello World

Output appears directly below the cell.

Important Jupyter Notebook Shortcuts

Shortcut	Function
Shift + Enter	Run current cell
Alt + Enter	Run and create new cell
A	Insert cell above
B	Insert cell below
D D	Delete current cell
Ctrl + S	Save notebook

Step 9: Create More Programs

Program 2

```
print("My Name is Krupa")
```

Output:

My Name is Krupa

Program 3

```
print("Welcome to Python Programming")
print("We are using VS Code")
```

Output:

Welcome to Python Programming
We are using VS Code

```
# This is my first Python program
```

```
print("Hello World")
```

Python ignores comments.